

Didactic PCB V1.0 user guide

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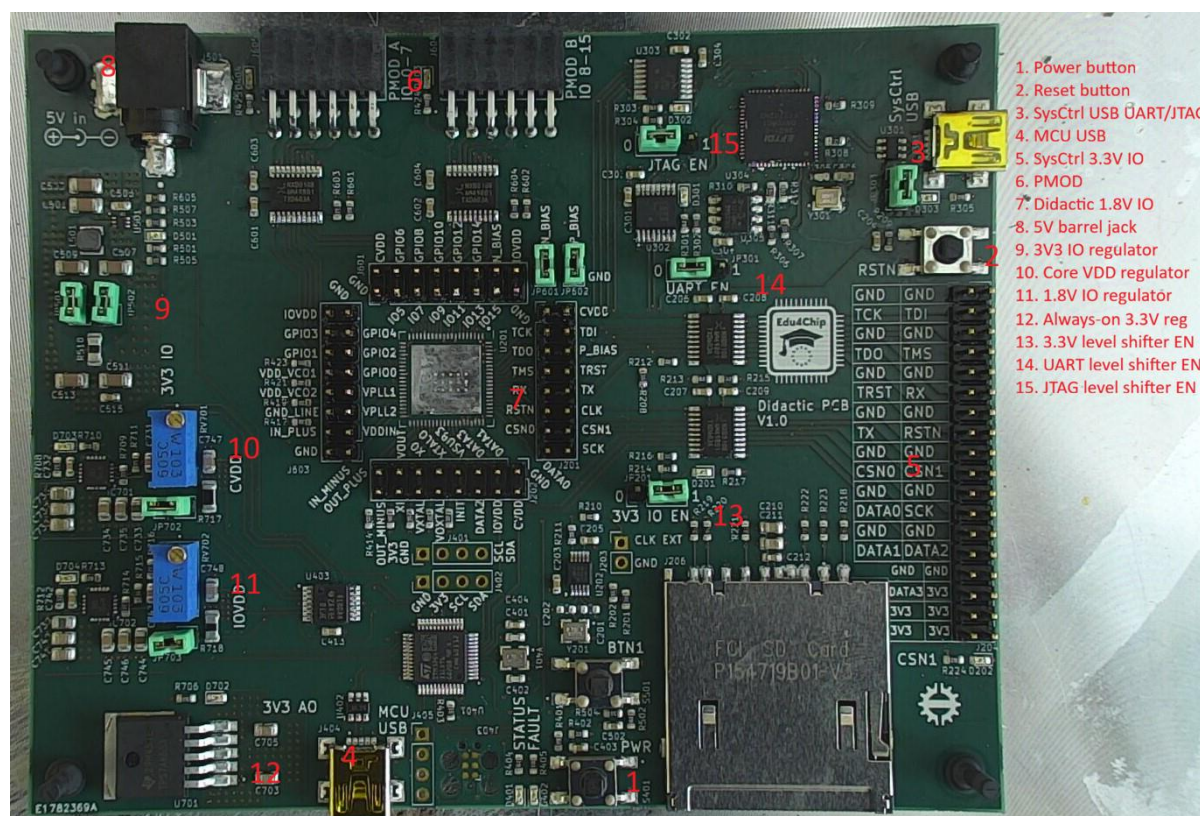


Figure 1. Didactic PCB

1.0 Power and start-up

The Didactic PCB requires external 5V source to function. The 5V can be inserted from a 5.5 mm x 2.1 mm barrel jack or from the SysCtrl Mini-USB port. Only one of these methods should be used at once. If barrel jack is used and the SysCtrl Mini-USB ports are used at the same time, then JP303 should be opened to disconnect the 5V power from the Mini-USB port.

After 5V power has been provided to the board, an LED (D303) near the SysCtrl Mini-USB port should be lit to indicate that power is being provided. LED (D702) should also be lit to indicate that always-on 3.3 V is being regulated. Finally, the Status LED (D401) should slowly blink to indicate that the microcontroller on the board is ready, and the rest of the board can be powered by pressing the PWR button (S401). After pressing the Power button, the microcontroller will enable 3.3V IO, 1.8V IO and 0.9V Core power rails and start the clock generator and finally release the reset for the Didactic chip. Successful start-up is indicated by the STATUS LED being lit and not blinking. An LED near each power rail's source will also be lit to indicate that the power rails are enabled. All STATUS LED patterns are shown in table 1.

LED pattern	Pattern meaning
Slowly blinking	Board is powered, but Didactic power is off
Rapid blinking	Didactic power is on, but fault monitoring is disabled
Fully on	Didactic is powered

Table 1. STATUS LED patterns

The power rails are always monitored for failure. The power rails will fail if their voltage differs too much from the set amount or if too much current is being drawn from them. All power rails are immediately powered down if a failure is noticed in any of the power rails. Failure is indicated by blinking the Fault LED (D402). The failed power rail is indicated by the blinking pattern of the Fault LED (table 2). After power rail failure, the board can be repowered by pressing the PWR button.

Blink amount	Failed power rail
1	Core
2	IO
3	3.3V IO

Table 2. Fault LED blinking patterns

If needed, the power regulators can be disconnected from the power rails by removing the jumpers near them (JP703, JP702, JP501 and JP502). This may be needed at the first bring-up process for each PCB as the power rail voltages may be set incorrectly. If the Didactic chip is assembled on the PCB before bring-up, the power regulators must be disconnected and adjusted.

Voltage of the IO and Core voltage rails can be adjusted using trimmers RV702 and RV701.

2.0 IO

2.1 SysCtrl 3.3V IO

All SysCtrl IOs are level shifted to 3.3V and available on header J204. The signals are level shifted using bidirectional auto direction sensing level shifters. The level shifters must be enabled using the 3V3 IO EN jumper. Because of a design mistake, the level shifters are enabled when the jumper is set to 0 and disabled when the jumper is set to 1. LED D201 near the jumper will indicate when the level shifters are enabled. Level shifted SPI signals are also connected to an SD card socket.

JTAG and UART signals are also level shifted using separate unidirectional level shifters and connected to FT2232HQ USB converter. These level shifters can be enabled with UART EN (JP301) and JTAG EN (JP302) jumpers. Pinout for the FT2232HQ is shown in table 3.

FT2232HQ pin	Net
ADBUS0	TCK
ADBUS1	TDI
ADBUS2	TDO
ADBUS3	TMS
BDBUS0	TX (Didactic RX)
BDBUS1	RX (Didactic TX)

Table 3. FT2232HQ pinout

2.2 Didactic 1.8V IO

All of Didactic 1.8V IO is directly accessible on pin headers J201, J202, J601 and J603. All analog pins 32-50 of Didactic are connected to ground using 0-ohm resistors. N BIAS and P BIAS are connected to ground using JP601 and JP602 jumpers.

2.3 PMOD

GPIO signals 0-15 of the Didactic are connected to two PMOD (PMOD A and PMOD B) connectors and level shifted to 3.3V using bidirectional auto direction sensing level shifters. These level shifters are always enabled. PMOD connect IO signal pinouts are shown in table 4.

PMOD A		PMOD B	
Didactic	PMOD	Didactic	PMOD
GPIO0	IO1	GPIO8	IO1
GPIO1	IO2	GPIO9	IO2
GPIO2	IO3	GPIO10	IO3
GPIO3	IO4	GPIO11	IO4
GPIO4	IO5	GPIO12	IO5
GPIO5	IO6	GPIO13	IO6
GPIO6	IO7	GPIO14	IO7
GPIO7	IO8	GPIO15	IO8

Table 4. PMOD pinout

3.0 Power monitor MCU

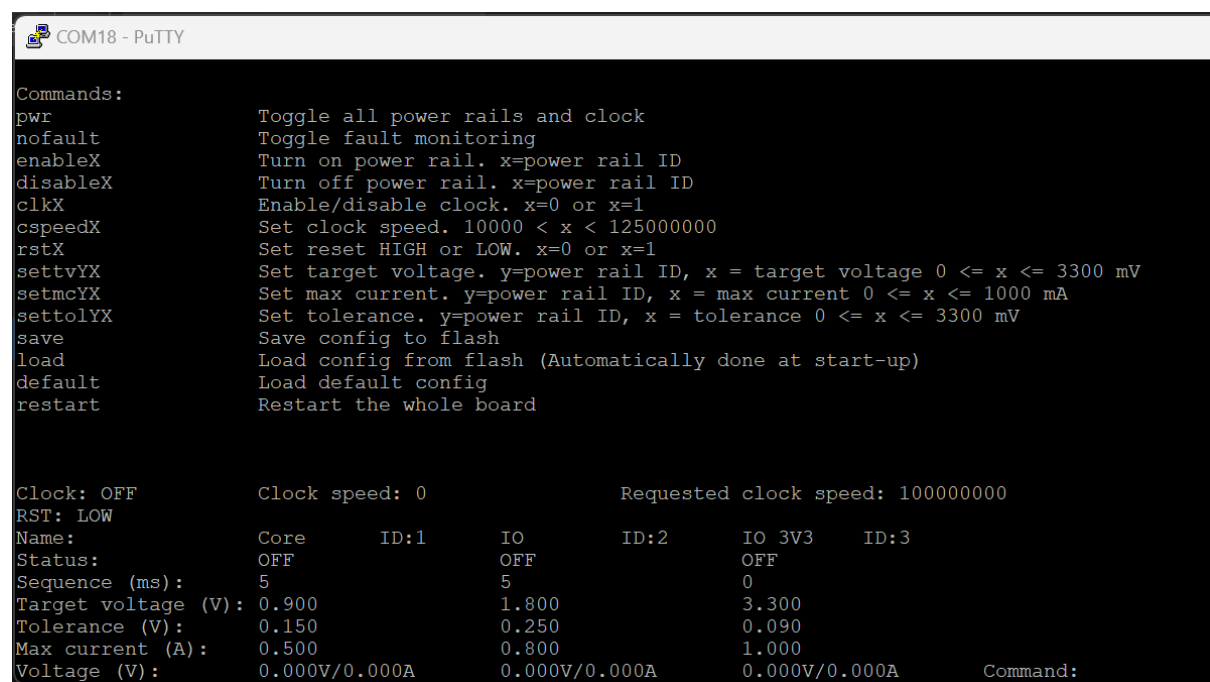
Microcontroller on the board is used to monitor and sequence the power rails and control the clock generator. The Didactic can be started by pressing the PWR button or the start-up process can be controlled using the microcontrollers virtual COM port. The COM port can be accessed by connecting a Mini-USB cable to the MCU USB port.

3.1 Virtual COM port

Any virtual terminal software can be used to connect to Virtual COM port. Baud rate can also be freely chosen.

The microcontroller will continuously print the voltage and current of all the power rails. The virtual terminal window should be made big enough so that a full line of text can fit without line changes. Commands can be typed into the virtual terminal, and all commands must be ended by pressing enter. Pressing enter will also always reprint the command list

Virtual terminal view is shown in figure 2. Configuration for all power rails is shown at the bottom of the terminal and the bottom most line is updated three times a second and it shows the current voltage and power consumption of the power rails.



```
COM18 - PuTTY

Commands:
pwr                Toggle all power rails and clock
nofault            Toggle fault monitoring
enableX            Turn on power rail. x=power rail ID
disableX           Turn off power rail. x=power rail ID
clkX               Enable/disable clock. x=0 or x=1
cspeedX            Set clock speed. 10000 < x < 125000000
rstX               Set reset HIGH or LOW. x=0 or x=1
settvYX            Set target voltage. y=power rail ID, x = target voltage 0 <= x <= 3300 mV
setmcYX            Set max current. y=power rail ID, x = max current 0 <= x <= 1000 mA
settolYX           Set tolerance. y=power rail ID, x = tolerance 0 <= x <= 3300 mV
save               Save config to flash
load               Load config from flash (Automatically done at start-up)
default            Load default config
restart            Restart the whole board

Clock: OFF          Clock speed: 0          Requested clock speed: 100000000
RST: LOW
Name:               Core      ID:1      IO        ID:2      IO 3V3    ID:3
Status:             OFF       OFF      OFF       OFF
Sequence (ms):      5        5        0
Target voltage (V): 0.900     1.800    3.300
Tolerance (V):      0.150     0.250    0.090
Max current (A):    0.500     0.800    1.000
Voltage (V):        0.000V/0.000A  0.000V/0.000A  0.000V/0.000A  Command:
```

Figure 2. Power management MCU virtual terminal view

3.2 Commands

Power rails and the clock can be controlled by writing commands to the virtual terminal. All supported commands are shown in table 5.

Command	Parameters	Explanation
pwr		Turns on/off all the power rails and clock. Same as pressing the PWR button
nofault		Disables/enables fault checking for all power rails. Status LED will blink quickly when power is enabled and fault checking is disabled
enableX	X = power rail ID	Enabled power rail ¹
disableX	X = power rail ID	Disable power rail ¹
clkX	X = 0 or X = 1	Enabled or disable clock. 1 = enable
cspeed	10000 < X < 125000000	Set requested clock speed. clk1 command must be sent after this command to enable clock with requested speed
rstX	X = 0 or X = 1	Set Didactic reset LOW or HIGH, LOW = reset
settvYX	Y = power rail ID, 0 mV <= X <= 3300 mV	Set power rail's target voltage
setmcYX	Y = power rail ID, 0 mA <= X <= 1000 mA	Set power rail's maximum current
settolYX	Y = power rail ID, 0 mV <= X <= 3300 mV	Set power rail's tolerance to voltage difference between target and actual voltage
save		Save current config to Flash
load		Load config from flash
default		Load default config
restart		Shutdown all power rails and restart the MCU

Table 5. Power management MCU commands

1. IO and Core power rails source their power from 3.3V IO power rails, so they cannot be powered before 3.3V IO power rails. IO and Core can be enabled before 3.3V IO power rails but they will be powered only after 3.3V IO has been powered.

3.3 Programming

The microcontroller can be programmed using SWD interface. This can be done using 4-pin pin header (J405) or 6-pin TC2030 Tag-Connect cable.